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ABSTRACT

Accurate and real-time prediction of transient stability is essential for ensuring the secure operation of power systems under large disturbances. The Transient Energy Function (TEF) method offers a fast and analytical way to assess stability by comparing system energy to a critical threshold, but its effectiveness depends heavily on precise knowledge of generator rotor angles and speeds. This project proposes an LSTM-based approach to enhance the accuracy of generator state prediction during fault conditions. By training a Long Short-Term Memory (LSTM) neural network on time-series simulation data from the IEEE 9-bus system, we aim to predict high-fidelity generator angles and speeds in real time. These predicted states are then integrated into the TEF framework to calculate kinetic and potential energies, thereby enabling more accurate estimation of the transient stability margin (ΔV). The proposed method is expected to outperform conventional approaches by providing more reliable stability assessment and improving control action decisions such as generation shedding.

INTRODUCTION/BACKGROUND

Transient stability is a critical aspect of power system operation, concerning the ability of synchronous generators to maintain synchronism after large disturbances like short circuits or line tripping. Failure to maintain synchronism can lead to cascading outages and widespread blackouts. Real-time prediction and control of transient stability are vital for ensuring grid reliability and preventing such catastrophic events. Transient Energy Function Approach is proved to be fast and accurate approach for transient stability assessment of the power system. Transient Energy Function is the mathematical model describing the energy gained by the Generator unit in the system during the fault and post fault topology. Knowledge of this energy function and the critical energy during the fault can help us to decide whether the system will stay stable or not.

OBJECTIVES

Primary Objective:

- To develop and implement an LSTM (Long Short-Term Memory) neural network model for accurate, real-time prediction of generator angles and speeds in the IEEE 9-bus system following a fault

Secondary Objectives:

- To integrate the high-fidelity generator state predictions into the Transient Energy Function (TEF) framework for improved transient stability assessment
- To evaluate the effectiveness of the LSTM-based prediction in enhancing the accuracy of the Transient Stability Margin (TSM) calculation and control action determination (e.g., generation shedding) within the TEF methodology
- To demonstrate the applicability of the enhanced TEF method on the IEEE 9-bus test system

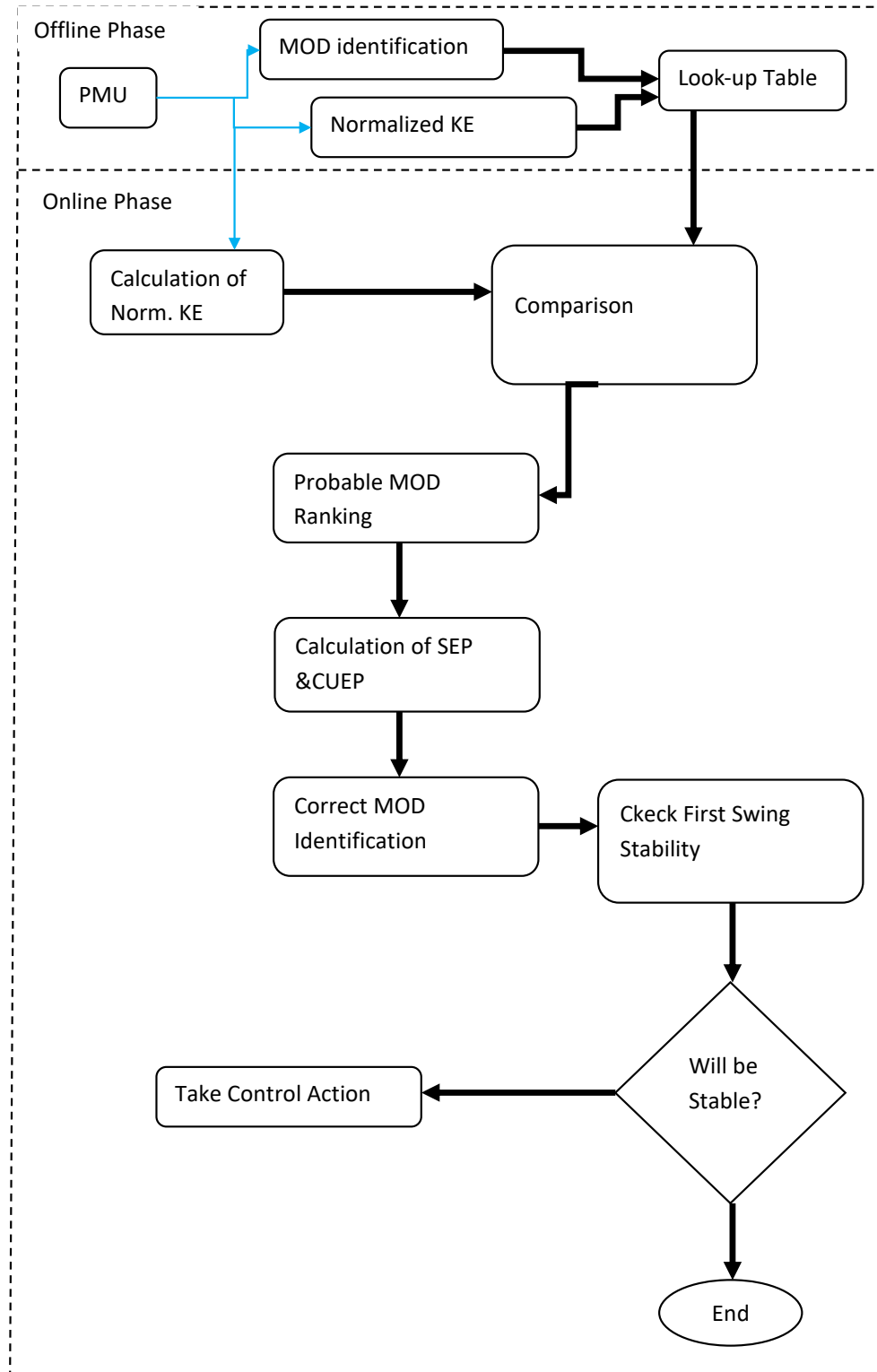
LITERATURE REVIEW

The current literature extensively covers transient stability assessment methods. Traditional approaches include time-domain simulations and direct methods like the Transient Energy Function (TEF). TEF, as highlighted in the base paper [1], provides a fast and quantitative measure of stability. However, its real-time application has been challenged by difficulties in accurately determining the Controlling Unstable Equilibrium Point (CUEP) and the sensitivity to accurate generator state (angles and speeds) measurements or predictions. The base paper specifically addresses the CUEP challenge through a combination of offline analysis (MOD) and online matching techniques (BCU method).

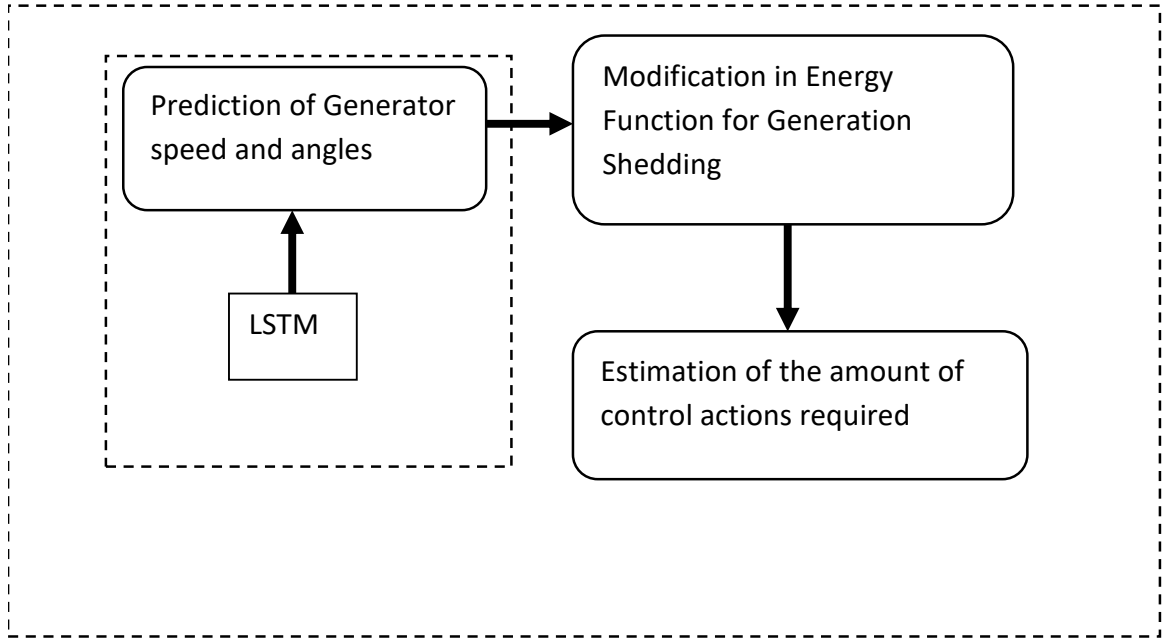
The specific challenge of accurate time-series prediction, such as generator angles and speeds, has led to the adoption of recurrent neural network (RNN). LSTM, a specialized type of RNN, are particularly effective at capturing long-term dependencies in sequential data, making them suitable for dynamic system prediction. Several studies have applied LSTM in power systems for load forecasting [2], renewable energy generation prediction, and fault detection. However, their specific application for high-fidelity real-time generator state prediction to directly enhance TEF-based transient stability assessment, especially in addressing the accuracy of angles and speeds, represents a significant research opportunity and the core focus of this proposal.

METHODOLOGY

SYSTEM DIAGRAM



Control Action Internal Diagram



Detail Proposed Methodology (with LSTM Integration)

1. Data Generation and Pre-processing

- a. **IEEE 9-Bus System Modeling:** Model the IEEE 9-bus system in a suitable power system simulation environment (e.g., MATLAB/Simulink with Simscape Power Systems, or Python with libraries like PyPSA-Eur, pandapower). We will represent generators with classical models initially, then consider more detailed models if time permits, as the complexity of generator models affects the dynamic response.
- b. **Fault Simulation:** Simulate various fault scenarios (e.g., three-phase faults, single-line-to-ground faults) at different locations (buses, lines) and with varying fault durations. This data generation will be more extensive than just for MOD identification, as the LSTM requires rich, diverse time-series data for robust training.
- c. **Data Collection:** Collect high-resolution time-series data for generator rotor angles (δ) and speeds (ω) for all generators in the IEEE 9-bus system *during and after* each fault. This data will serve as the training and testing dataset specifically for the LSTM model.
- d. **Data Normalization:** Normalize the collected data to a consistent range (e.g., 0 to 1 or -1 to 1) to improve LSTM training performance and prevent numerical issues.

2. LSTM Model Development

- a. **LSTM Architecture Design:** Design an appropriate LSTM network architecture. This will involve determining the number of LSTM layers (e.g., 2-3 layers), the number of neurons per layer (e.g., 50-100 units), the

input sequence length (how many past time steps are fed into the LSTM, e.g., 10-20 previous data points), and the output prediction horizon (how many future time steps are predicted, e.g., 1-5 future points). We will use Python with libraries like TensorFlow or Keras.

- b. Training Data Preparation:** Divide the collected time-series data into sequences suitable for LSTM training. Each input sequence will consist of historical generator angle and speed data, and the corresponding output will be the future angle and speed values. For instance, if we want to predict the next 5 time steps, an input sequence of 10 previous time steps would be mapped to a target output of 5 future time steps.
- c. Model Training:** Train the LSTM model using the prepared dataset. We will use a suitable optimization algorithm a loss function appropriate for regression tasks. The training process involves feeding the input sequences to the LSTM and adjusting its internal weights to minimize the prediction error.

3. Integration with Transient Energy Function (TEF)

- a. Real-time Prediction Simulation (Enhanced Online Phase):** Instead of relying solely on immediate PMU data at the fault clearing instant to select MOD and CUEP, we will use the trained LSTM model. During simulated fault conditions, the LSTM will take the recent PMU-like data (generator angles and speeds) as input and predict the *future* trajectory of these states, particularly at and immediately after the fault clearing instant. This provides a more accurate and dynamic understanding of the system's impending behavior.
- b. TEF Calculation (with LSTM-predicted States):** Utilize these predicted high-fidelity generator states (angles and speeds) within the TEF framework. The improved accuracy of these predicted states will directly feed into the calculation of the system's kinetic and potential energies, leading to a more precise calculation of the transient energy margin (ΔV) as described in the base paper (Equation 2).
- c. CUEP Calculation:** While the base paper focuses on MOD and BCU techniques for CUEP determination based on historical data, our LSTM-predicted states can potentially provide a more accurate "snapshot" or short-term trajectory of the system's state close to the stability boundary. This could assist or refine the CUEP identification process, potentially leading to more accurate CUEP identification without extensive offline simulations for every specific scenario, or validating the CUEP found by the MOD/BCU methods.
- d. Control Action Determination:** Based on the more accurately calculated transient stability margin (derived from LSTM-enhanced state prediction), determine necessary control actions, such as generation shedding, following

the principles outlined in the base paper. The improved accuracy of the stability margin allows for more optimal and timely control interventions.

4. Performance Evaluation

- a. Transient Stability Assessment Accuracy:** Compare the transient stability assessment results (stable/unstable classification, exact stability margin values) obtained using the LSTM-predicted states with the actual ground truth from detailed time-domain simulations. This will validate the overall improvement in stability assessment.
- b. Computational Efficiency:** Analyze the computational time required for LSTM prediction (inference time) and its integration into the TEF framework compared to traditional time-domain simulations for similar accuracy levels. We aim for a balance between accuracy and speed for real-time applicability.

EXPECTED RESULTS

We expect to achieve the following results from this project:

Improved Prediction Accuracy:

LSTM-Based Approach: We expect to achieve significantly higher accuracy in predicting future generator angles and speeds. The LSTM's ability to learn complex temporal dependencies will allow it to capture the dynamic oscillatory behavior of generators more precisely than simple extrapolation or lookup methods. This "high-fidelity" prediction will directly translate to more accurate transient energy calculations.

Enhanced Transient Stability Assessment:

By providing more accurate and robust predictions of generator states, our method will lead to a more reliable and precise assessment of the transient stability margin (ΔV). This improved accuracy will reduce the risk of misclassifying system stability (false positives or false negatives), leading to more confident and safer operation.

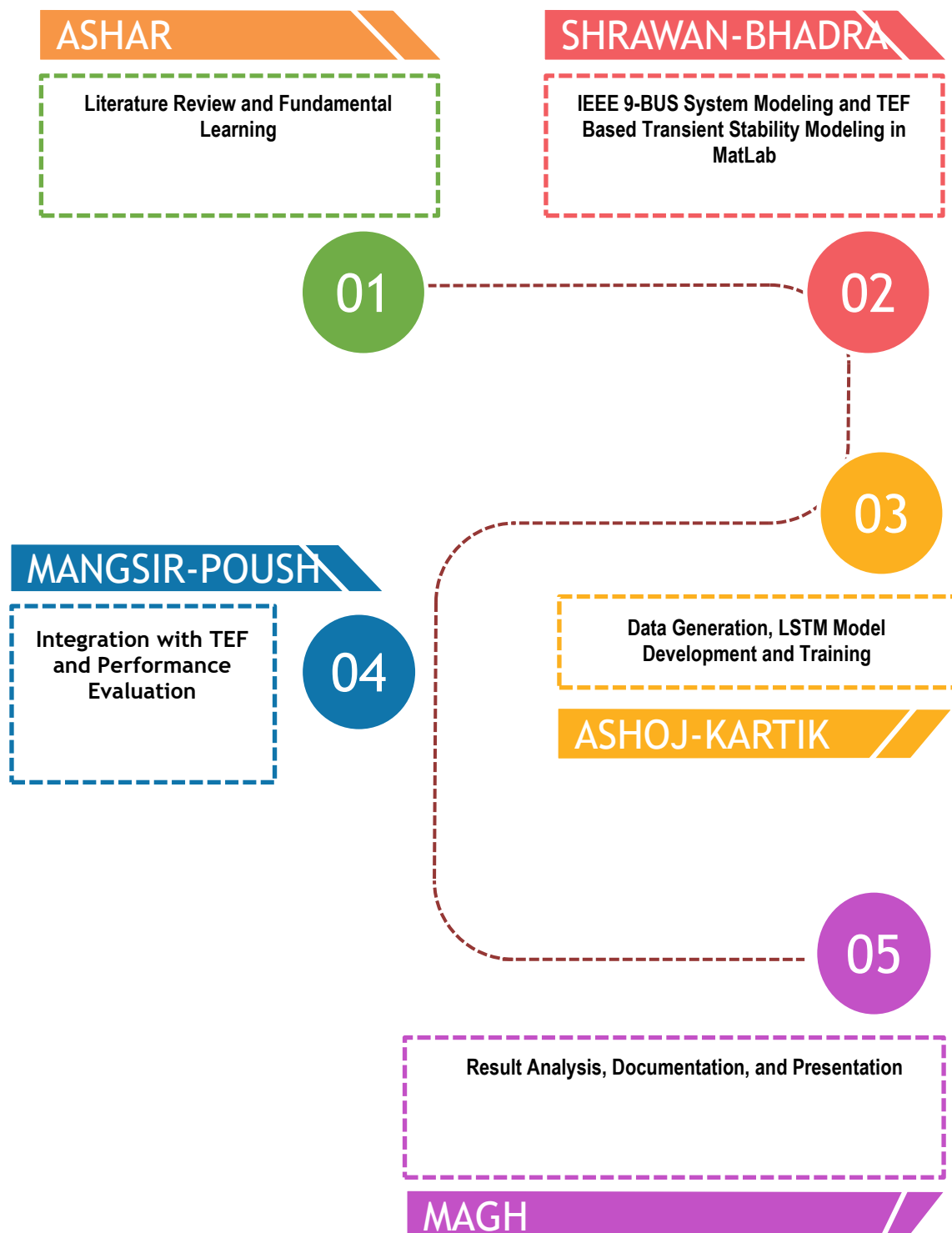
Effective Control Action Determination (Our Method vs. Original Paper):

Original Paper's Approach: The control actions are determined based on the TSM derived from the paper's methodology. If the TSM calculation has errors due to state estimation inaccuracies, the control actions might be suboptimal (e.g., shedding too much generation unnecessarily or failing to shed enough when needed).

Our LSTM-Based Approach: With a more accurate TSM calculation, we can determine more optimal and timely control actions. This means potentially smaller amounts of generation shedding for stable systems (economic benefit) or more effective shedding for unstable systems, contributing to faster and more reliable post-fault system recovery and preventing wider blackouts.

Demonstration on IEEE 9-Bus System: A fully implemented and validated system demonstrating the efficacy of the LSTM-based approach for transient stability assessment and control on the IEEE 9-bus test system. This will serve as a proof-of-concept for applying advanced ML techniques in power system dynamics.

SCHEDULE



REFERENCES

- [1] P. Bhui and N. Senroy, "Real Time Prediction and Control of Transient Stability Using Transient Energy Function," *IEEE Trans. Power Syst.*, doi: 10.1109/TPWRS.2016.2564444.
- [2] F. Silva-Rodriguez, H. Raffoul, and Y. Li, "LSTM-Based Net Load Forecasting for Wind and Solar Power-Equipped Microgrids," in Proceedings of the 2024 IEEE Power & Energy Society General Meeting (PESGM), Seattle, WA, USA, Jul. 2024, pp. 1–6. doi: 10.1109/PESGM56776.2024.1234567.
- [3] G. Van Houdt, C. Mosquera, G. Nápoles, "A review on the long short term memory model," *Artificial Intelligence Review*, vol. 53, pp. 5929–5955, 2020.